

Project 1 : Personal Portfolio Website Deployment

Aim

To create a personal portfolio website using HTML and CSS and deploy it on **Amazon S3 Static Website Hosting**.

Objective

The objective of this experiment is to design a personal portfolio website, upload it to an Amazon S3 bucket, configure the bucket for static website hosting, make the website publicly accessible, and access it through the S3 website endpoint.

Tools and Technologies Used

- HTML5
 - CSS3
 - Visual Studio Code
 - Git
 - GitHub
 - Amazon Web Services (AWS)
 - Amazon S3
 - IAM
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AWS Services Used

- Amazon S3
 - Static Website Hosting
 - IAM (Identity and Access Management)
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Git Concepts Used

- git init
- git add
- git commit
- git branch
- git remote
- git push

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Git Commands Used

```
git init
git add .
git commit -m "Initial Portfolio Website"
git branch -M main
git remote add origin https://github.com/username/portfolio.git
git push -u origin main
```

Procedure

Step 1: Create the Portfolio Website

1. Create a folder named **Portfolio**.
 2. Open the folder in Visual Studio Code.
 3. Create the index.html file.
 4. Write the HTML and CSS code for the portfolio website.
 5. Save the file.
 6. Open the HTML file in a web browser and verify that the website is displayed correctly.
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Step 2: Create a GitHub Repository

1. Log in to GitHub.
 2. Create a new repository named **portfolio**.
 3. Keep the repository public.
 4. Do not initialize it with a README file.
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Step 3: Initialize Git

1. Open Git Bash in the project folder.
2. Run:

```
git init
```

3. Add the project files.

```
git add .
```

4. Commit the project.

```
git commit -m "Initial Portfolio Website"
```

5. Rename the default branch.

```
git branch -M main
```

6. Connect the GitHub repository.

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```
git remote add origin https://deepakkatwa.github.io/Portfolio/
```

7. Push the files.

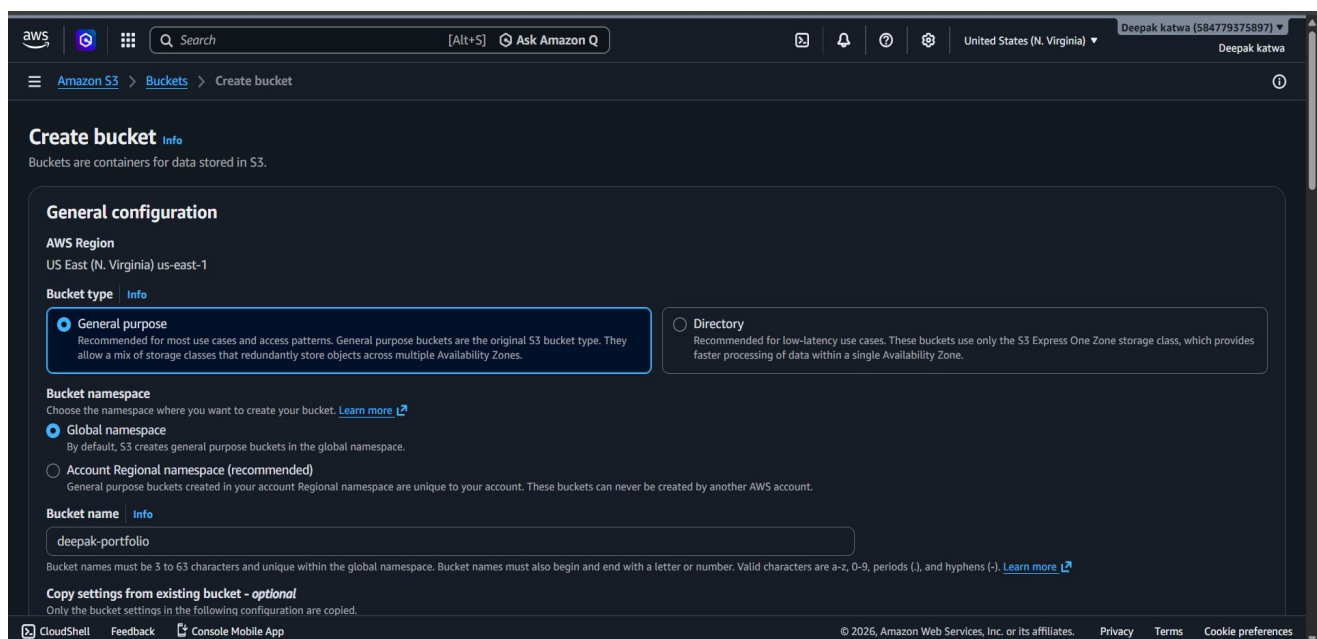
```
git push -u origin main
```

Step 4: Login to AWS

1. Open the AWS Management Console.
2. Sign in using your AWS account.
3. Search for **Amazon S3**.

Step 5: Create an S3 Bucket

1. Click **Create Bucket**.
2. Enter a globally unique bucket name.
3. Select the desired AWS Region.
4. Uncheck **Block all public access**.
5. Acknowledge the warning.
6. Click **Create Bucket**.



Step 6: Upload Website Files

1. Open the created bucket.
2. Click **Upload**.
3. Select the following files:
 - **index.html**
 - Click **Upload**.

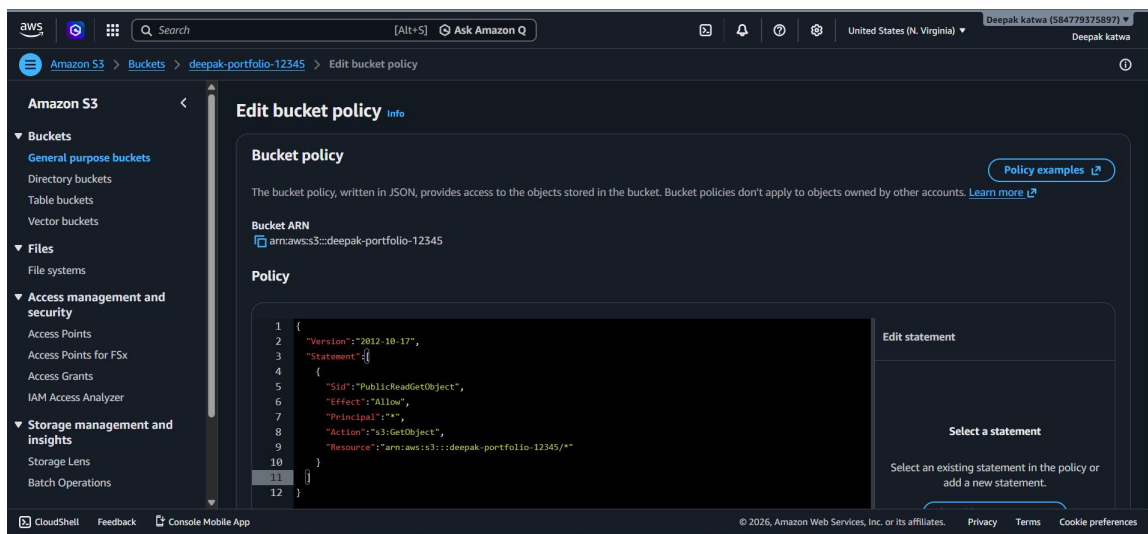
Step 7: Enable Static Website Hosting

1. Open the bucket.
2. Go to the **Properties** tab.
3. Scroll to **Static Website Hosting**.
4. Click **Edit**.
5. Enable Static Website Hosting.
6. Choose **Host a Static Website**.
7. Enter:
 - **Index Document:** `index.html`
 - **Error Document:** `error.html` (optional)
8. Save the changes.

Step 8: Configure Bucket Policy

1. Open the **Permissions** tab.
2. Edit the Bucket Policy.
3. Paste the following policy:
4.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::deepak-portfolio-12345/*"
    }
  ]
}
```
5. Save the policy.

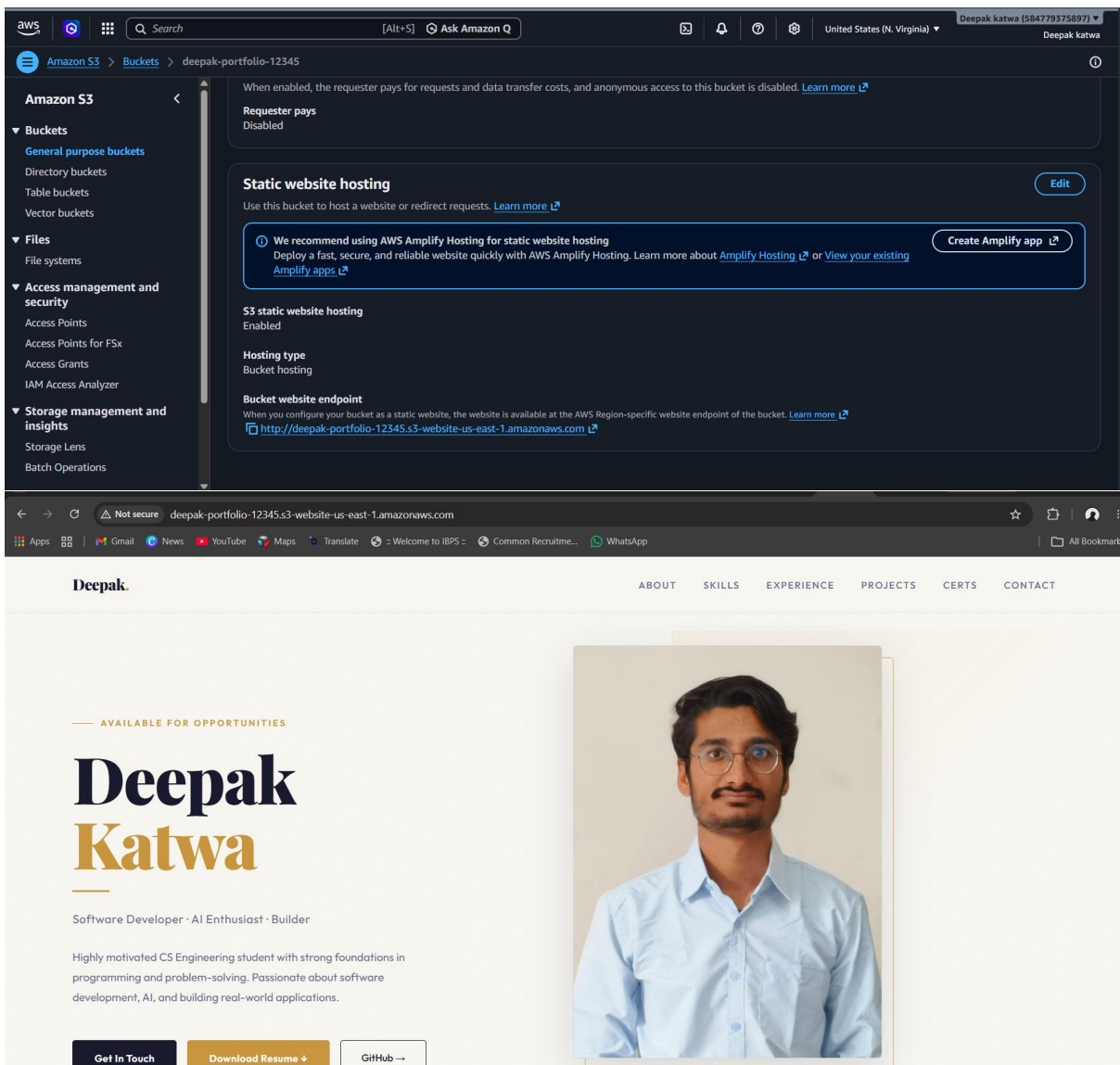


Step 9: Access the Website

1. Open the **Properties** tab.
2. Scroll to **Static Website Hosting**.
3. Copy the **Bucket Website Endpoint**.
4. Open the endpoint URL in a web browser to view the deployed portfolio website.

Expected Output

The portfolio website is successfully hosted on Amazon S3 and can be accessed using the Static Website Endpoint URL from any device with internet access.



Result

A personal portfolio website was successfully created using HTML and deployed on **Amazon S3 Static Website Hosting**. The website was made publicly accessible using a bucket policy and can be accessed through the S3 Static Website Endpoint URL.